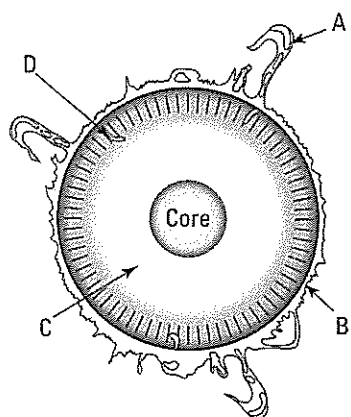


The sun and nuclear fusion

Student: Class:

The structure of the sun

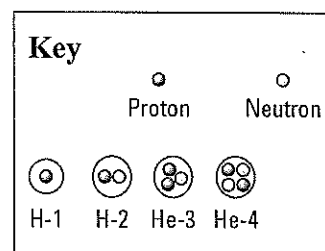
1. Our sun has a layered structure. Use the diagram and table below to match the letters (A–D) to the correct structure.



Description of structure	Letter
1. Radiation zone — this layer is outside the core; energy transfers to this layer from the core	
2. Solar prominence — these structures extend up to 500 000 km into space	
3. Convection zone — convection currents carry energy towards the photosphere	
4. Photosphere — light is emitted into space from this surface	

Nuclear fusion

2. The sun generates energy by nuclear fusion. During this process hydrogen nuclei are joined together to form helium nuclei. This process occurs in three steps. Use the key to construct diagrams for each step. The first step has been done for you.



Step 1: Two hydrogen (H-1) nuclei fuse to form a deuterium (H-2) nucleus.



Step 2: One deuterium nucleus and one hydrogen nucleus fuse to form a helium-3 (He-3) nucleus. A gamma ray is also formed.

Step 3: Two helium-3 nuclei fuse to form a helium-4 (He-4) nucleus. Two protons and large amounts of energy are released in this step.